LN7 Portfolio Dynamics

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Outlines

Textbook: Arbitrage Theory in Continuous Time: Chapter 6

1. Introduction
2. Self-financing Portfolios

Notation

Definition 1

$N =$ *the number of different types of stocks*
$h_i(t) =$ *number of shares of type* i *held during the period* $[t, t + \Delta t)$.
$h(t) =$ *the portfolio* $[h_1(t), \cdots, h_N(t)]$ *held during period* t.
$c(t) =$ *the amount of money spent on consumption per unit time during the period* $[t, t + \Delta t)$.
$S_i(t) =$ *the price of one share of type* i *during the period* $[t, t + \Delta t)$.
$V(t) =$ *the value of the portfolio* h *at time* t.

We will consider so called self-financing portfolio. In a situation without any consumption ($c(t) = 0$) we have the following V-dynamics for self-financing portfolio.

$$dV(t) = h(t)dS(t)$$

Derivation of the V-dynamics

Mathematically,

$$
\begin{aligned}
h(t-\Delta t)S(t) &= h(t)S(t) + c(t)\Delta t \\
S(t)\Delta h(t) + c(t)\Delta t &= 0 \\
S(t)\Delta h(t) - S(t-\Delta t)\Delta h(t) + S(t-\Delta t)\Delta h(t) + c(t)\Delta t &= 0 \\
\Delta S(t)\Delta h(t) + S(t-\Delta t)\Delta h(t) + c(t)\Delta t &= 0 \\
dS(t)dh(t) + S(t)dh(t) + c(t)dt &= 0
\end{aligned}
$$

From $V(t) = h(t)S(t)$,

$$
\begin{aligned}
dV(t) &= h(t)dS(t) + S(t)dh(t) + dh(t)dS(t) \\
&= h(t)dS(t) - c(t)dt \\
dV(t) &= h(t)dS(t)
\end{aligned}
$$

Mathematical Definition of the Central Concepts

Definition 2

Let the N-dimensional price process $\{S(t) : t \geq 0\}$ *be given.*

1. *A* **portfolio strategy** *(most often simply called a portfolio) is any* $\mathcal{F}_t^S$*-adapted N-dimensional process* $\{h(t); t \geq 0\}$.
2. *The portfolio h is said to be* **Markovian** *if it is of the form*

$$h(t) = h(t, S(t)),$$

   for some function $h : R_+ \times R^N \to R^N$.
3. *The* **value process** V^h *corresponding to the portfolio h is given by*

$$V^h(t) = \sum_{i=1}^{N} h_i(t) S_i(t).$$

Mathematical Definition of the Central Concepts Continued

Definition 2 Continued

4. A **consumption process** is any $\mathcal{F}_t^S$-adapted one-dimensional process $\{c(t); t \geq 0\}$.
5. A portfolio–consumption pair (h, c) is called **self-financing** if the value process V^h satisfies the condition

$$dV^h(t) = h(t)dS(t) - c(t)dt.$$

Relative Portfolio

Definition 3

For a given portfolio h the corresponding **relative portfolio** *u is given by*

$$u_i(t) = \frac{h_i(t)S_i(t)}{V^h(t)}, i = 1, \cdots, N,$$

where we have

$$\sum_{i=1}^{N} u_i(t) = 1.$$

Relative Portfolio

Lemma 1

A portfolio–consumption pair (h, c) is self-financing if and only if

$$dV^h(t) = V^h(t) \sum_{i=1}^{N} u_i(t) \frac{dS_i(t)}{S_i(t)} - c(t)dt.$$

Value Process

Lemma 2

Let c be a consumption process, and assume that there exist a scalar process Z and a vector process $q = (q_1, \cdots, q_N)$ such that

$$dZ(t) = Z(t) \sum_{i=1}^{N} q_i(t) \frac{dS_i(t)}{S_i(t)} - c(t)dt.$$

Now define a portfolio h by

$$h_i(t) = \frac{q_i(t)Z(t)}{S_i(t)}.$$

Then the value process V^h is given by $V^h = Z$, the pair (h, c) is self-financing, and the corresponding relative portfolio u is given by $u = q$.

Proof of Lemma 2

Proof

$$
\begin{aligned}
V^h(t) &= \sum_{i=1}^{N} h_i(t) S_i(t) \\
&= \sum_{i=1}^{N} q_i(t) Z(t) \\
&= Z(t).
\end{aligned}
$$